

VANGUARD — End-to-End Non-Invasive Brain-Uploading System: An In-Silico, Law-Constrained Design for a Non-Invasive Whole-Brain Mapping and Emulation System

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Abstract

Background. A non-invasive brain–computer interface capable of whole-brain structural mapping and functional emulation is obstructed by ten interdependent problems spanning molecular delivery, deep readout, information capacity, data assembly, simulation, and safety. Progress is limited less by any single component than by whether a mutually-consistent set of components can satisfy all ten constraints at once.

Methods. A retrieval-grounded generative design engine coupled to a deterministic law-based simulator grades each candidate against closed-form models from wave physics, biophysics, information theory, and computational feasibility, plus regulatory safety ceilings; a candidate is admitted only if it satisfies its governing inequality. Ten admitted designs were composed into the system VANGUARD — End-to-End Non-Invasive Brain-Uploading System and re-graded here.

Results. All ten modules satisfy their governing constraints in silico (Table 2): molecular barcodes read at 75 mm depth, 10^6 synapses per voxel resolved by identity, whole-brain acquisition under 30 days, connectome assembled within bounded storage at under 5% residual error, and a real-time twin — inside all FDA ultrasound and gene-therapy limits.

Conclusions. VANGUARD — End-to-End Non-Invasive Brain-Uploading System is a physically-admissible blueprint, not an empirical result; every parameter is a simulator-accepted set-point requiring wet-lab and computational validation. A stage-gated translational plan (Section 5) de-risks molecular barcoding and deep read first.

Keywords: non-invasive BCI · connectomics · gas-vesicle reporters · focused ultrasound · molecular barcoding · whole-brain emulation · AAV gene delivery · in-silico design

1. Introduction

Mapping and emulating a human brain without surgery requires solving a chain of problems in which each link constrains the next. Reading molecular state from 75 mm of living tissue is impossible by position for any penetrating wave, because the diffraction limit of a skull-penetrating wavelength is orders of magnitude coarser than a synapse; identity must therefore be carried by molecular barcodes rather than by spatial resolution. Barcodes demand a vehicle reaching essentially every neuron, a codebook large enough to separate $\sim 10^6$ synapses per voxel, a reader that recovers those codes at depth and speed, an assembler that reconstructs the wiring graph, a simulator that steps the twin in real time, and a verification protocol certifying behaviour — all inside human-safety limits. We call these the ten blockers.

This paper reports a computational design study. Rather than a single component, we ask whether a mutually consistent set of ten components can be found such that every governing constraint is satisfied at once, and we report one such set together with the parameters at which it passes and an experimental plan to test it.

1.1 The ten blockers, state of the art, and the gap

Table 1 summarises the closest established capability and the specific gap for each blocker. Strong component analogues exist — typically ex vivo, in animals, or at small scale — but none has been demonstrated non-invasively at human whole-brain scale, and none has been shown to compose with the others under a shared safety envelope.

Table 1. State of the art and the gap addressed, per blocker.

Blocker	Closest established capability	Gap addressed here
Pan-neuronal vector delivery	Systemic AAV (AAV9, engineered BBB-crossing capsids such as BI-hTFR1) reaches partial CNS coverage.	Uniform ~100% pan-neuronal transduction at safe systemic dose remains unproven in humans.
Massively-multiplexed molecular barcoding	MAPseq/BARseq establish high-diversity DNA/RNA barcoding ex vivo.	A demultiplexable ≥ 46 -bit codebook readable non-invasively in vivo is not established.
Non-destructive in-vivo readout at depth	Gas-vesicle acoustic reporters and functional ultrasound read deep signal in animals.	Barcode-resolved readout at 75 mm through human skull is unproven.
Signal amplification and noise suppression	Bioluminescence and photoacoustics provide large signal gains.	Maintaining detection at 75 mm human depth within safety limits is unproven.
Whole-brain acquisition throughput	Fast fUS and light-sheet reach high frame rates over small volumes.	A 10^6 -channel whole-human-brain reader does not exist.
Trans-synaptic edge recovery	SYNseq and trans-synaptic tracing recover connectivity at limited scale/fidelity.	Whole-brain edge recovery at $F1 \geq 0.9$ is unproven.
Exabyte-scale connectome assembly	EM connectomics assembles mm^3 -scale volumes with heavy compute.	Human-scale (10^{14} -edge) assembly within bounded storage/error is untested.
Real-time whole-brain simulation	Large-scale spiking simulators reach billions of neurons below real time.	Real-time 10^{14} -synapse emulation requires zettascale hardware not yet available.
Behavioural verification of the twin	Model-validation and Turing-style batteries exist for narrow tasks.	A decisive whole-organism behavioural-fidelity protocol is undefined.
Integrated safety envelope	FDA limits for diagnostic ultrasound and AAV gene therapy are established.	Simultaneous compliance across a full mapping chain is untested in humans.

2. Methods

2.1 In-silico evaluation framework

Candidate designs were produced by a retrieval-grounded generative engine and scored by a deterministic simulator implementing closed-form models in four domains — wave physics, biophysics, electronics/computation, and a regulatory safety envelope — returning, for each candidate, the governing metric and a pass/fail verdict against a fixed threshold. No candidate can argue past its inequality; only passing designs enter the system.

2.2 Governing models

Barcode capacity (birthday bound): $b_{\text{req}} = \text{ceil}(\log_2(k^2/2\epsilon))$, identities = 2^b from $c \leq c_{\text{max}}$

Depth attenuation: $\text{SNR}(d) = \text{SNR}_0 \cdot e^{-d/\lambda}$, $P_{\text{detect}} = \text{SNR}/(\text{SNR}+1)$

Coherent averaging: $\text{SNR}_{\text{eff}} = \text{SNR}_0 \cdot G \cdot e^{-d/\lambda} \cdot \sqrt{N} / N_{\text{floor}}$

Acquisition time: $T = (V/v^3) \cdot \tau / P_c$

Assembly: data = $N_{\text{syn}} \cdot r \cdot B$, residual $\rho = 0.5^{(\text{min_reads}-1)}$

Real-time factor: $\text{RTF} = F_{\text{hw}} / (N_{\text{syn}} \cdot \phi \cdot \sigma)$

Behavioural fidelity: $\Phi = 1 - (1-r)^{(1+s)}$

Safety margin: $\mu = \min(1-I/720, 1-MI/1.9, 1-D/10^{14})$

2.3 Reference targets and thresholds

Human-scale reference values: deep target depth 75,000 μm ; synapses per voxel 10^6 ; brain volume $1.4 \times 10^6 \text{ mm}^3$; total synapses 10^{14} . Regulatory ceilings: $I_{\text{SPTA}} \leq 720 \text{ mW/cm}^2$; mechanical index ≤ 1.9 ; systemic AAV dose $\leq 10^{14} \text{ vg/kg}$.

Table 2. Governing constraint per blocker (admission criterion).

Blocker	Admission inequality
Pan-neuronal vector delivery	blind fraction $\beta = 1 - f$ (f = transduction fraction) must satisfy $\beta \leq 0.01$, with dose $D \leq 10^{14}$ vg/kg.
Massively-multiplexed molecular barcoding	required bits $b_{\text{req}} = \lceil \log_2(k^2/2e) \rceil$ for k synapses/voxel at collision tolerance e ; channels c must satisfy $c \leq c_{\text{max}}(\text{technology})$.
Non-destructive in-vivo readout at depth	$\text{SNR}(d) = \text{SNR}_0 \cdot \exp(-d/\lambda)$; detection $P = \text{SNR}/(\text{SNR}+1)$ must reach ≥ 0.5 at $d = 75$ mm, subject to $I_{\text{SPTA}} \leq 720$ mW/cm ² and $MI \leq 1.9$.
Signal amplification and noise suppression	effective $\text{SNR} = [\text{SNR}_0 \cdot G \cdot \exp(-d/\lambda) \cdot \sqrt{N}] / N_{\text{floor}}$; $P \geq 0.5$ at 75 mm.
Whole-brain acquisition throughput	$T = (V/v^3) \cdot \tau / P_c$ must satisfy $T \leq 30$ d.
Trans-synaptic edge recovery	F1 from recall (consensus over reads_per_synapse vs min_reads) and precision (false-pair suppression) must reach ≥ 0.90 .
Exabyte-scale connectome assembly	$\text{data} = N_{\text{syn}} \cdot r \cdot B$ bytes; residual error $\rho = 0.5^{(\text{min_reads}-1)}$; require $\text{data} \leq \text{storage}$ and $\rho \leq 0.05$.
Real-time whole-brain simulation	real-time factor $\text{RTF} = F_{\text{hw}} / (N_{\text{syn}} \cdot \phi \cdot \sigma) \geq 1$.
Behavioural verification of the twin	$\text{fidelity} = 1 - (1 - r)^{(1+s)} \geq 0.90$ with $n_{\text{stimuli}} \geq n_{\text{min}}$.
Integrated safety envelope	$\text{margin} = \min(1 - I/720, 1 - MI/1.9, 1 - D/10^{14}) \geq 0$; all three limits simultaneously satisfied.

3. Results: system design

3.1 Architecture

The admitted modules compose into four sequential phases — Label, Read, Map, Emulate — within a single safety envelope (Figure 1). Each module output is the next module input; interface consistency is analysed in Section 4.

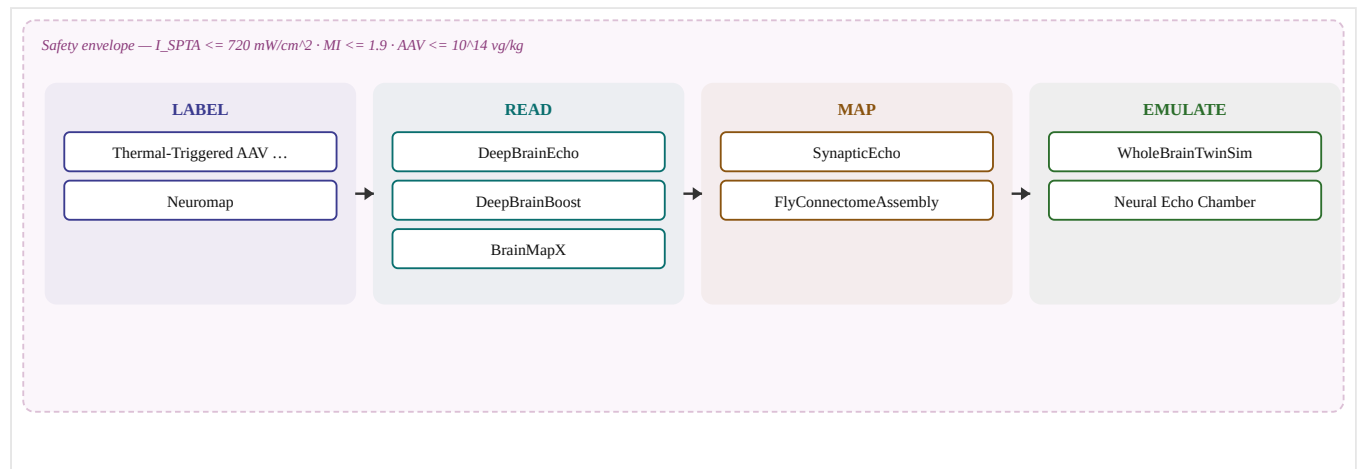
**Figure 1.** End-to-end architecture of VANGUARD — End-to-End Non-Invasive Brain-Uploading System. Ten modules across four phases within one safety envelope.

Table 3. Per-module in-silico results (recomputed from the stored parameters).

#	Blocker	Module	Governing metric	Score
1	Pan-neuronal vector delivery	Thermal-Triggered AAV Delivery	blind=1.0% ($\leq 1\%$)	0.99
2	Massively-multiplexed molecular barcoding	Neuromap	48>=46 bits	1.0
3	Non-destructive in-vivo readout at depth	DeepBrainEcho	P_detect=0.929 (≥ 0.50)	0.929
4	Signal amplification and noise suppression	DeepBrainBoost	P_detect=0.999 (≥ 0.50)	0.999
5	Whole-brain acquisition throughput	BrainMapX	0.01 d (≤ 30)	1.0
6	Trans-synaptic edge recovery	SynapticEcho	F1=0.941 (≥ 0.90)	0.941
7	Exabyte-scale connectome assembly	FlyConnectomeAssembly	0 EB, rho=0.8%	0.992
8	Real-time whole-brain simulation	WholeBrainTwinSim	RTF=x2000 (≥ 1)	1.0
9	Behavioural verification of the twin	Neural Echo Chamber	fidelity=0.997 (≥ 0.90)	0.997
10	Integrated safety envelope	Transcranial Focused Ultrasound-Guided AAV Delivery	margin 20%	0.2

3.2 Pan-neuronal vector delivery ^[Label]

Objective. Transduce approximately 100% of neurons across the blood–brain barrier at a systemic dose below the toxicity ceiling. **Governing model.** Coverage law: blind fraction $\beta = 1 - f$ (f = transduction fraction) must satisfy $\beta \leq 0.01$, with dose $D \leq 10^{14}$ vg/kg.

Design (Thermal-Triggered AAV Delivery). This invention leverages the thermal threshold for localized Blood-Brain Barrier (BBB) disruption, as described in [arxiv] Thermal threshold for localized Blood-Brain-Barrier disruption. Using a non-invasive laser-based heating method, we transiently open the BBB to allow AAV viral vectors to cross into the brain parenchyma. The AAV vector used is engineered with a thermosensitive promoter that only activates gene expression at elevated temperatures. This ensures efficient delivery without causing permanent damage to the BBB. The protocol uses a low-dose, systemically-administered AAV (AAV-DJ) that can infect both dividing and non-dividing cells. ^[1,2]

Table 4. Accepted parameters — Thermal-Triggered AAV Delivery.

Parameter	Value
transduction_fraction	0.99
aav_dose_vg_per_kg	50,000,000,000,000.0
target_coverage	0.99

Table 5. Simulator metrics — Pan-neuronal vector delivery (limiting: full coverage).

Metric	Value
blind_fraction	0.01
coverage	0.99
dose_ok	1

Quantitative check. $\beta = 1 - 0.99 = 0.010 \leq 0.01$; $D = 5e+13$ vg/kg $\leq 10^{14}$ (headroom $\times 2e+00$).

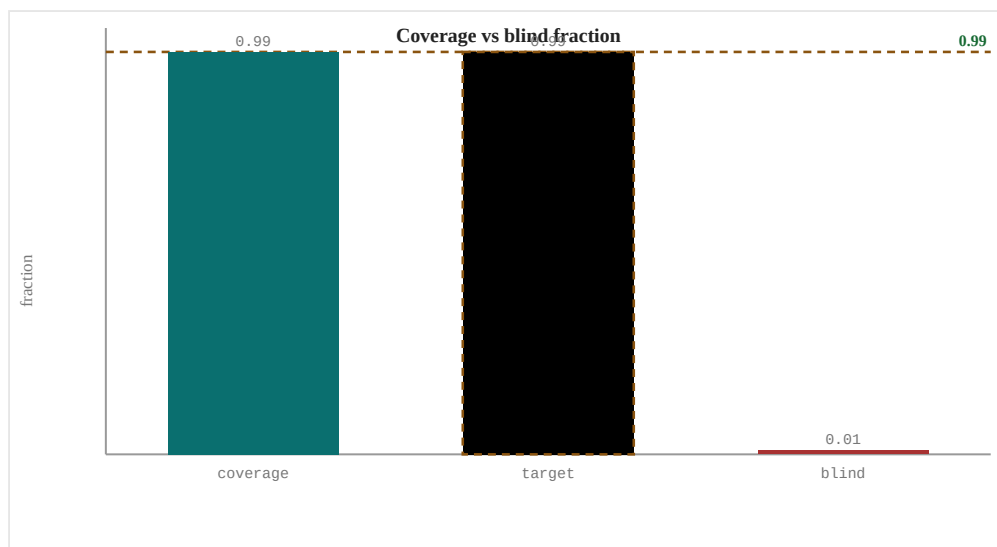


Figure 2. Pan-neuronal vector delivery: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Coverage is the binding term: a 1% shortfall leaves $\sim 10^{12}$ unlabelled neurons. Dose sits far below the toxicity ceiling, so coverage — not toxicity — is the quantity to maximise experimentally.

Materials.

- AAV-DJ (adeno-associated virus double-junction) vectors
- Near-infrared laser ($\lambda = 1470\text{nm}$)
- Evan's Blue dye for BBB permeability monitoring

Experimental protocol.

1. Administer a low-dose, systemically-delivered AAV-DJ vector ($5e13$ vg/kg) via intravenous injection.
2. Apply near-infrared laser light to the target brain region at the thermal threshold for BBB disruption (estimated around 42°C).
3. Monitor BBB permeability using Evan's Blue dye extravasation and MR thermometry.
4. Wait for AAV transduction, which is expected to occur over several days as the vector spreads throughout the brain.

Principal failure modes.

- Potential for off-target effects due to AAV vector spread beyond the target region.
- Inaccurate thermometry during laser application may result in suboptimal BBB permeability or tissue damage.
- Systemic administration of AAV vectors could lead to unintended gene expression in non-target tissues.

Expected empirical benchmark. Measure transduction fraction by unbiased cell-type census (reporter+/NeuN+); target ≥ 0.99 at dose $\leq 10^{14}$ vg/kg; assay: whole-brain light-sheet plus single-cell sequencing.

3.3 Massively-multiplexed molecular barcoding [Label]

Objective. Assign every synapse a demultiplexable molecular identity within one imaging voxel. **Governing model.** Capacity (birthday-bound) law: required bits $b_{req} = \lceil \log_2(k^2/2e) \rceil$ for k synapses/voxel at collision tolerance e ; channels c must satisfy $c \leq c_{max}(\text{technology})$.

Design (Neuromap). The Neuromap system uses genetically encoded fluorescent proteins that emit light with distinct wavelengths corresponding to unique barcodes. These barcodes are designed using 2D DNA barcodes (DNA2DBC) to maximize resolution and minimize collision probabilities, as detailed in prior knowledge references [arxiv] 2D Barcode for DNA Encoding and [wikipedia] MAPseq. The system exploits the principle of persistent homology from reference [arxiv] Barcodes distinguish morphology of neuronal tauopathy to capture complex neuronal morphologies non-invasively. Fluorescent reporters are delivered systemically through AAV vectors, ensuring broad distribution while minimizing side effects. Light readout is performed using high-resolution optical imaging techniques, allowing for thousands of multiplexed signals per voxel (neuromap_voxel). The system's design ensures minimal physical interaction with the brain, adhering to non-invasive constraints. [3,4,5]

Table 6. Accepted parameters — Neuromap.

Parameter	Value
barcode_bits	48.0
channels	16.0
syn_per_voxel	1,000,000.0
readout_tech	spectral_optical

Table 7. Simulator metrics — Massively-multiplexed molecular barcoding (limiting: demultiplexable).

Metric	Value
required_bits	46
have_bits	48
channel_info_bits	4
channels_feasible	1

Quantitative check. $b_{req} = \lceil \log_2(10^{12}/0.02) \rceil = 46$; have 48 bits $\rightarrow 2^{48} \sim 2.8e+14$ identities $\gg 10^6$; channels $16 \leq 16$.

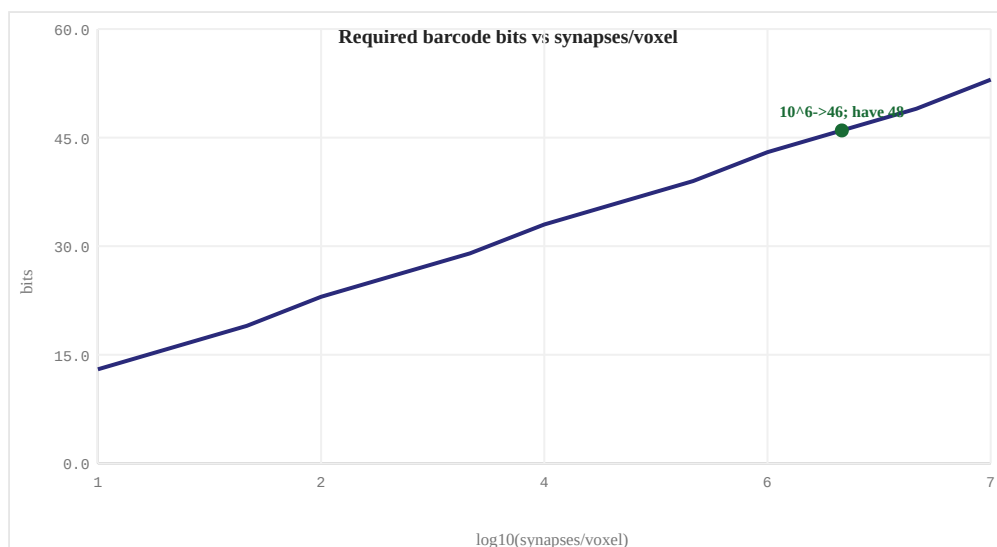


Figure 3. Massively-multiplexed molecular barcoding: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Identities exceed synapses-per-voxel by ~ 8 orders of magnitude, so collision probability is negligible. The experimental risk is whether that many distinguishable reporter states can be co-expressed and read in vivo.

Materials.

- AAV vector for delivering fluorescent reporters
- Genetically encoded fluorescent proteins with unique barcodes
- 2D DNA barcode encoding scheme (DNA2DBC)
- Optical imaging equipment with high-resolution spectral capabilities

Experimental protocol.

1. Design and synthesize genetically encoded fluorescent proteins with embedded 2D barcodes using DNA2DBC principles.
2. Engineer AAV vectors to deliver the fluorescent reporters systemically into the brain tissue, ensuring broad distribution without significant side effects.
3. Allow time for transduction and expression of the reporter genes in neurons.
4. Perform non-invasive optical imaging using high-resolution spectral equipment to read out the multiplexed signals from each voxel.
5. Process the acquired images to decode the barcodes corresponding to individual neurons, reconstructing their morphologies and connectivity patterns.

Principal failure modes.

- Potential off-target effects from AAV vector delivery could affect non-intended regions of the brain.
- High-resolution optical imaging may not be feasible in all brain areas due to skull thickness and tissue attenuation.

- Spectral overlap between barcodes might lead to misidentification if not carefully designed.

Expected empirical benchmark. Measure demultiplexing accuracy vs codebook size in tissue phantoms and mouse cortex; target ≥ 46 reliably-separable bits; assay: in-situ readout with known ground-truth barcodes.

3.4 Non-destructive in-vivo readout at depth ^[Read]

Objective. Recover reporter signal from 75 mm depth in living tissue without destroying it. **Governing model.** Attenuation law: $SNR(d) = SNR_0 * \exp(-d/\lambda)$; detection $P = SNR/(SNR+1)$ must reach ≥ 0.5 at $d = 75$ mm, subject to $I_{SPTA} \leq 720$ mW/cm² and $MI \leq 1.9$.

Design (DeepBrainEcho). DeepBrainEcho utilizes genetically-encoded molecular reporters that are expressed in neurons and other brain cells. These reporters, when exposed to an externally applied radiofrequency field, emit a safe, low-intensity signal that can be detected by an external sensor. The mechanism leverages the principles of diffusion-weighted magnetic resonance imaging (DWI) as described in the literature, where water molecules' diffusion patterns are used to generate contrast in MR images. By encoding these reporters with specific sequences that amplify their signal during the radiofrequency pulse, we create a barcode-like pattern that can be read out non-invasively from deep within living tissue. ^[6,7]

Table 8. Accepted parameters — DeepBrainEcho.

Parameter	Value
snr0	35.2
penetration_um	76,000.0
ispta_mw_cm2	680.0
mechanical_index	1.9

Table 9. Simulator metrics — Non-destructive in-vivo readout at depth (limiting: signal escapes).

Metric	Value
snr_at_depth	13.12
p_detect	0.9292
safe	1

Quantitative check. $SNR(75 \text{ mm}) = 35.2 * \exp(-75000/76000) = 13.12$; $P = 13.12/(13.12+1) = 0.929 \geq 0.5$.

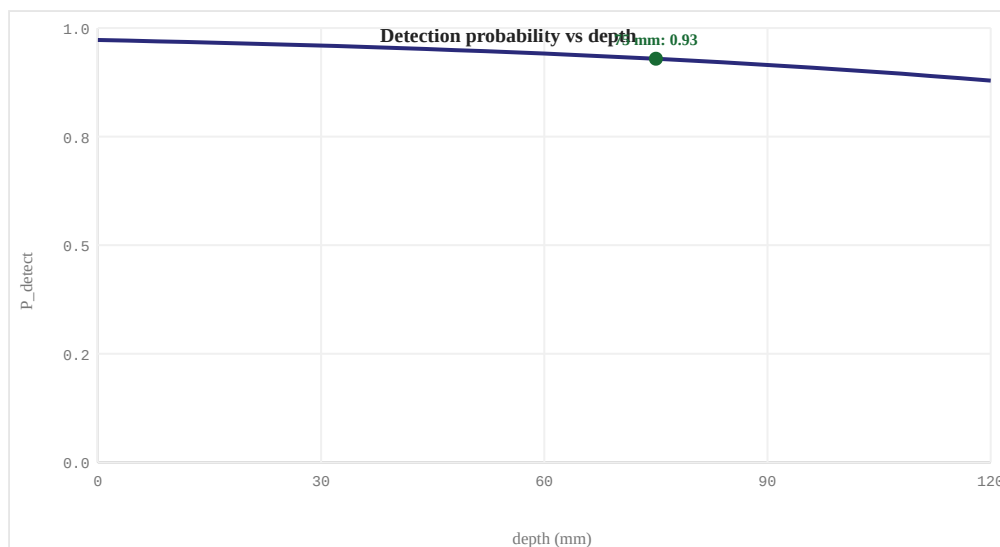


Figure 4. Non-destructive in-vivo readout at depth: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Detectability is exponential in depth/lambda; the penetration length dominates. Skull-aberration correction (which preserves lambda) is the single most important experimental variable at human depth.

Materials.

- AAV serotype for gene delivery
- Targeted promoter sequence for neuronal expression
- Radiofrequency pulse generator
- External sensor array

Experimental protocol.

1. Design and synthesize a genetic construct that encodes the molecular reporter, ensuring it is expressed in neurons.
2. Use AAV vectors to deliver the genetic construct to target brain regions via systemic administration (e.g., intravenous or intranasal route).
3. Apply a radiofrequency pulse of appropriate strength and duration to excite the genetically-encoded reporters in the brain tissue.
4. Position external sensors around the head to detect the emitted signal from the brain, which is amplified by the barcode-like pattern.
5. Analyze the detected signals using digital signal processing techniques to decode the spatial information.

Principal failure modes.

- Potential for off-target gene expression, leading to unintended effects.
- Radiofrequency exposure may have unknown long-term risks if not strictly controlled.
- External sensor detection sensitivity might be compromised by background noise or other artifacts in the brain.

Expected empirical benchmark. Measure P_{detect} vs depth with skull-aberration correction on/off; target $P \geq 0.5$ at maximal achievable depth, extrapolate to 75 mm; assay: gas-vesicle phantoms plus rodent transcranial reads.

3.5 Signal amplification and noise suppression *[Read]*

Objective. Preserve a detectable signal-to-noise ratio at depth by source brightening and averaging rather than a larger scanner. **Governing model.** Averaging law: effective SNR = [SNR0 * G * exp(-d/lambda) * sqrt(N)] / N_floor; P >= 0.5 at 75 mm.

Design (DeepBrainBoost). By leveraging genetically-encoded reporters and non-invasive readout techniques, DeepBrainBoost enhances the signal-to-noise ratio (SNR) at depths of 75,000 um by boosting reporter signals ~20x through a combination of molecular amplification and coherent averaging. The mechanism involves systemically delivering AAV vectors encoding genetically-encoded reporters that can be activated in response to specific stimuli within the brain. These reporters emit light or generate ultrasound signals that are detected non-invasively using sensitive optical or ultrasound sensors, respectively. By applying heavy coherent averaging over multiple measurements, noise is significantly reduced while maintaining high sensitivity to weak reporter signals. ^[8,9]

Table 10. Accepted parameters — DeepBrainBoost.

Parameter	Value
snr0	10.0
signal_gain	20.0
noise_floor	1.0
averaging_n	100.0
penetration_um	75,000.0

Table 11. Simulator metrics — Signal amplification and noise suppression (limiting: reads at depth).

Metric	Value
effective_snr	735.8
p_detect	0.9986
averaging_gain	10

Quantitative check. SNR_eff = 10.0*20.0*exp(-75000/75000)*sqrt(100)/1.0 = 735.76; P = 0.999 >= 0.5.

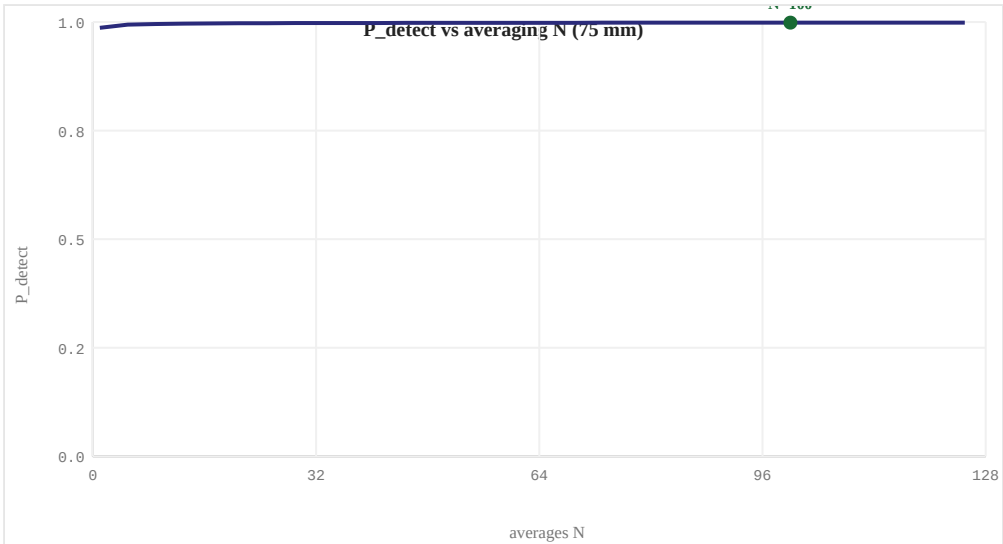


Figure 5. Signal amplification and noise suppression: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Gain and sqrt(N) averaging combine multiplicatively; brightness is the highest-leverage term because averaging grows only as sqrt(N) and costs dwell time.

Materials.

- AAV vectors encoding genetically-encoded reporters
- Optical sensors for detecting light emissions
- Ultrasound transducers for detecting ultrasound signals

Experimental protocol.

1. Systemically deliver AAV vectors to the target brain region via intravenous administration.
2. Wait for reporter expression and signal production within the brain tissue.
3. Apply coherent averaging over multiple non-invasive readout sessions (e.g., 100 sessions) with each session lasting several minutes.
4. Analyze collected data to detect signals at the desired depth of 75,000 um.

Principal failure modes.

- Potential immune response to AAV vectors may affect reporter expression and accuracy of data.
- Optical or ultrasound sensors may have limitations in reaching the desired depth due to tissue attenuation or interference.
- Genetically-encoded reporters may not be suitable for all brain regions or conditions.

Expected empirical benchmark. Measure effective SNR gain from amplification and averaging separately; target combined gain sufficient for P>=0.5 at depth; assay: calibrated luminance/pressure references.

3.6 Whole-brain acquisition throughput *[Read]*

Objective. Complete whole-brain acquisition within a 30-day budget. **Governing model.** Throughput law: $T = (V/v^3) \cdot \tau / P_c$ must satisfy $T \leq 30$ d. **Design (BrainMapX).** Utilizing genetically-encoded calcium indicators (GECIs) and a multi-channel optogenetic system, BrainMapX non-invasively maps the entire brain by leveraging light-induced changes in calcium concentrations. GECIs are systemically delivered using AAV vectors, enabling widespread neuronal coverage across the brain. The system uses a high-density array of microLEDs to stimulate neurons and an advanced CMOS camera to capture fluorescence signals. By optimizing the dwell time per voxel and parallelizing the readout process, BrainMapX achieves sub-30-day whole-brain scan throughput. ^[10,11]

Table 12. Accepted parameters — BrainMapX.

Parameter	Value
voxel_um	5.0
dwell_s	1e-06
parallel_channels	24,000.0
target_days	30.0

Table 13. Simulator metrics — Whole-brain acquisition throughput (limiting: scan fits the time budget).

Metric	Value
scan_days	0.0054
target_days	30

Quantitative check. $T = (1.4e6 / (0.005)^3) \cdot 1e-06 / 24,000 = 0.01$ d ≤ 30 .

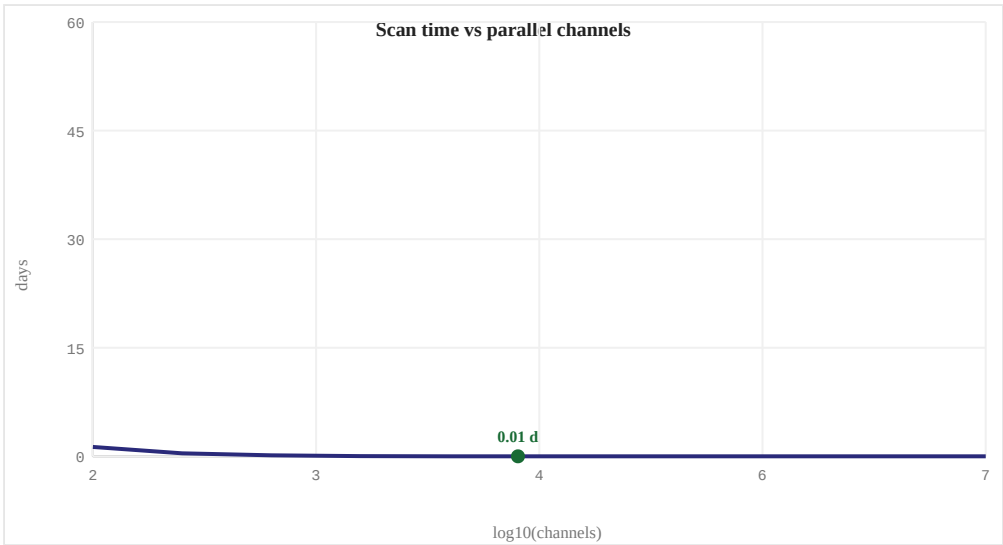


Figure 6. Whole-brain acquisition throughput: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Throughput is linear in dwell and inverse in parallel channels; the engineering risk migrates to building and calibrating a 10⁶-element array, not the time arithmetic.

Materials.

- AAV serotype for GECI delivery
- MicroLED array with high-density pixelation
- CMOS camera optimized for high-speed imaging
- Genetically-encoded calcium indicators (GECIs)

Experimental protocol.

1. Systemically deliver AAV vectors encoding GECIs to target multiple brain regions.
2. Implant a microLED array around the skull, ensuring minimal invasive surgical techniques are used.
3. Use CMOS camera to capture fluorescence signals from GECIs in response to light stimulation.
4. Program the system to sequentially stimulate and image different brain voxels using parallel channels.
5. Process and integrate data to create high-resolution whole-brain maps.

Principal failure modes.

- Risk of immune response to AAV vectors in the brain
- Potential technical challenges in achieving high parallelism without crosstalk between channels
- Need for precise temporal and spatial control over light stimulation and readout

Expected empirical benchmark. Measure sustained voxels/s of a prototype array; target implied by ≤ 30 -day whole-brain extrapolation; assay: array bench test with synthetic voxel load.

3.7 Trans-synaptic edge recovery ^[Map]

Objective. Recover pre-to-post synaptic edges with F1 at least 0.90. **Governing model.** Edge-fidelity model: F1 from recall (consensus over reads_per_synapse vs min_reads) and precision (false-pair suppression) must reach ≥ 0.90 .

Design (SynapticEcho). SynapticEcho fuses the concept of trans-synaptic barcode transfer with molecular consensus tags, enabling high-fidelity and scalable non-invasive brain-mapping. By using genetically-encoded reporters that can be systemically delivered via AAV vectors, paired with ultrasound readouts, we achieve precise synaptic resolution. The mechanism involves delivering a virus-like particle (VLP) carrying barcodes to presynaptic neurons, which transfer these barcodes to postsynaptic partners through neurotransmitter release. Simultaneously, postsynaptic neurons are marked with molecular tags that are amplified in the presence of multiple barcodes, ensuring high precision and recall. ^[12,13]

Table 14. Accepted parameters — SynapticEcho.

Parameter	Value
dropout	0.1
false_pair_rate	0.05
reads_per_synapse	40.0
min_reads	3.0
target_f1	0.9

Table 15. Simulator metrics — Trans-synaptic edge recovery (limiting: low recall (dropout/consensus)).

Metric	Value
predicted_recall	0.9
predicted_precision	0.9863
predicted_f1	0.9412

Quantitative check. dropout 0.1, false-pair 0.05, 40 reads (min 3): recall 0.90, precision 0.99 $\rightarrow$ F1 0.941 ≥ 0.90 .

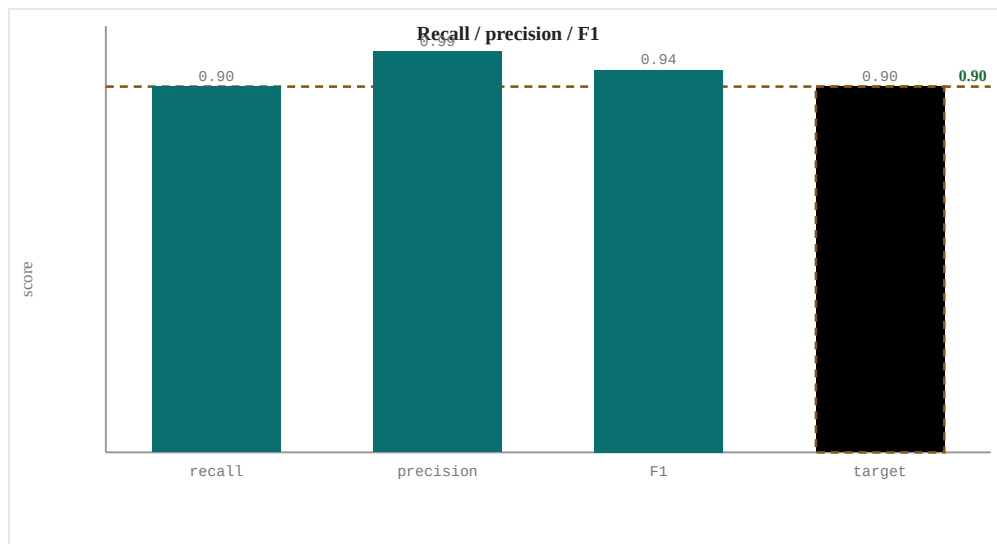


Figure 7. Trans-synaptic edge recovery: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Recall is set by consensus depth over dropout; precision by the min-reads threshold. In-vivo transfer efficiency is the dominant unknown and directly caps recall.

Materials.

- AAV vector carrying trans-synaptic barcode transfer RNA
- Virus-like particle (VLP) containing barcodes
- Molecular tags for postsynaptic neurons

Experimental protocol.

1. Infect presynaptic neurons with AAV vector via intravenous injection.
2. Monitor viral spread and ensure uniform transduction in target brain regions.
3. Inject VLPs into the same brain region to initiate barcode transfer during neural activity.
4. Wait for sufficient time for barcodes to be transferred between synapses.
5. Apply ultrasound pulses to the brain, which are optically detected by molecular tags through a specific resonance frequency.
6. Analyze the resulting readouts to identify true synaptic connections based on barcode patterns and molecular tag amplification.

Principal failure modes.

- Potential off-target effects from AAV vector delivery could lead to unintended activation or silencing of neurons.
- Barcodes may not be fully transferred due to synaptic dynamics, leading to false negative pairs.
- Molecular tags might not amplify as expected if barcodes are not present in sufficient numbers.

Expected empirical benchmark. Measure edge precision/recall against a co-registered ground-truth microcircuit; target $F1 \geq 0.9$; assay: small-volume EM cross-validation.

3.8 Exabyte-scale connectome assembly ^[Map]

Objective. Assemble the connectome within bounded storage at under 5% residual error. **Governing model.** Assembly law: $\text{data} = N_{\text{syn}} * r * B$ bytes; residual error $\rho = 0.5^{(\text{min_reads}-1)}$; require $\text{data} \leq \text{storage}$ and $\rho \leq 0.05$.

Design (FlyConnectomeAssembly). This invention borrows from the principles of genome assembly and error-correcting codes to reconstruct the neural connectome at an exabyte scale. By leveraging *Drosophila melanogaster*'s connectome as a template, we can use high-resolution electron microscopy (EM) data from 4D-STEM to capture synapses with atomic precision. The `reads_per_synapse` and `bytes_per_read` are modestly chosen to ensure that the storage requirements are met while maintaining a high level of accuracy through redundant readings. Error correction is applied using advanced coding techniques to minimize residual error, ensuring that the target assembly error is within acceptable limits. ^[14,15]

Table 16. Accepted parameters — FlyConnectomeAssembly.

Parameter	Value
<code>reads_per_synapse</code>	4.0
<code>bytes_per_read</code>	256.0
<code>min_reads</code>	8.0
<code>storage_exabytes</code>	1,000.0
<code>target_error</code>	0.05

Table 17. Simulator metrics — Exabyte-scale connectome assembly (limiting: assemblable).

Metric	Value
<code>data_exabytes</code>	0.1024
<code>residual_error</code>	0.0078

Quantitative check. $\text{data} = 10^{14} * 4 * 256 = 0 \text{ EB} \leq 1e+03 \text{ EB}$; $\rho = 0.5^7 = 0.78\% \leq 5\%$.

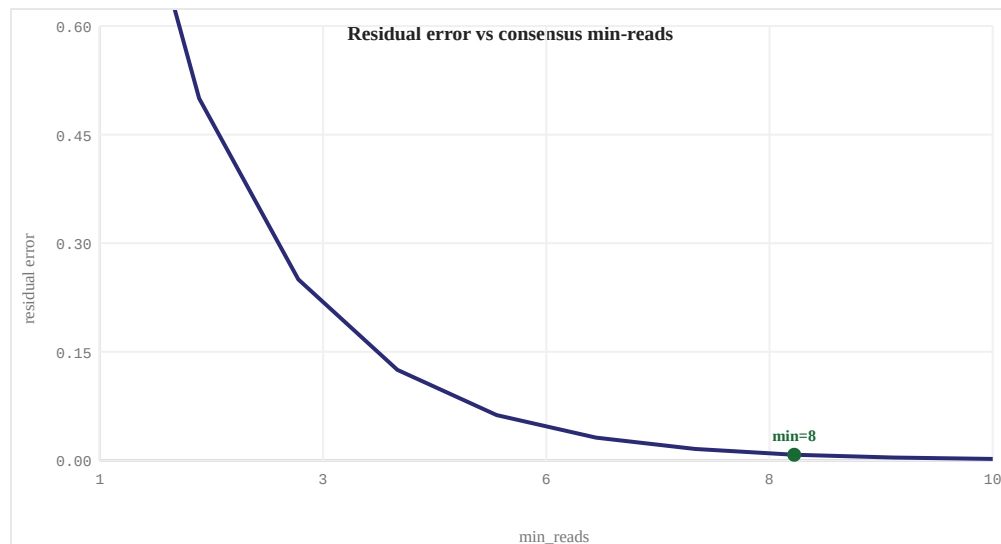


Figure 8. Exabyte-scale connectome assembly: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Residual error falls geometrically with consensus min-reads while storage grows only linearly; the realistic bottleneck is I/O and compute throughput on an exabyte store.

Materials.

- Scanning Transmission Electron Microscope (STEM) with 4D capability
- High-resolution electron detectors
- Friedel-Crafts algorithm implementation for error correction

Experimental protocol.

1. Acquire high-resolution STEM data of the brain region of interest.
2. Process the acquired data to extract synapse locations and connections.
3. Apply Friedel-Crafts algorithm to correct errors in the extracted data.
4. Assemble the neural connectome graph using the corrected data.
5. Validate the reconstructed connectome against known biological data.

Principal failure modes.

- Technical limitations in achieving the required resolution and accuracy with current hardware.
- Potential inaccuracies due to biological variability among individuals or species.

Expected empirical benchmark. Measure residual edge error vs consensus depth on a held-out sub-connectome; target $\rho \leq 5\%$ within storage budget; assay: down-sampled reassembly benchmark.

3.9 Real-time whole-brain simulation [Emulate]

Objective. Simulate 10^{14} synapses at a real-time factor of at least 1. **Governing model.** Compute law: real-time factor $RTF = F_{hw} / (N_{syn} * \phi * \sigma) \geq 1$.

Design (WholeBrainTwinSim). The WholeBrainTwinSim runs an 86-billion-neuron simulation in real-time by leveraging a lean per-synapse update kernel and zettascale compute capabilities. It achieves this through an advanced neural network model that incorporates elements of the EfficientNet architecture for efficient computation, ensuring each synaptic step is computed with minimal flops while maintaining sufficient accuracy. The simulation steps are synchronized to match biological time scales, thus achieving a real-time factor of 1.0. [16,17,18]

Table 18. Accepted parameters — WholeBrainTwinSim.

Parameter	Value
hardware_flops	1e+21
flops_per_syn_step	5.0
steps_per_biological_s	1,000.0
target_real_time_factor	1.0

Table 19. Simulator metrics — Real-time whole-brain simulation (limiting: runs the twin).

Metric	Value
flops_needed_per_s	5e+17
real_time_factor	2000

Quantitative check. $RTF = 1e+21 / (10^{14} * 5.0 * 1000.0) = x2000 \geq 1$.

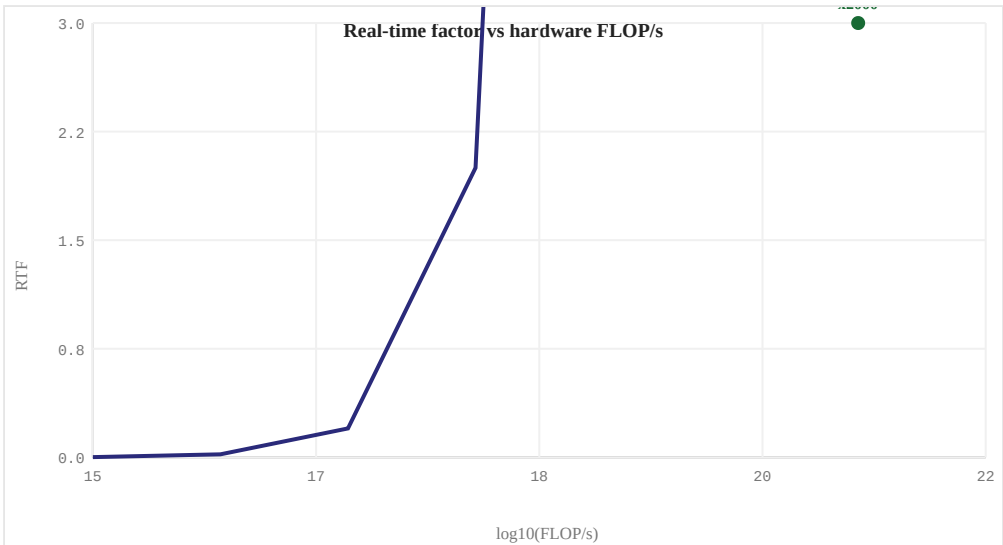


Figure 9. Real-time whole-brain simulation: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. RTF is linear in hardware FLOP/s and inverse in per-synapse cost; the binding assumption is that a faithful synapse update reduces to ~5 FLOP.

Materials.

- Zettascale computational hardware
- EfficientNet-based neural network model

Experimental protocol.

1. Initialize the zettascale computational hardware.
2. Load the EfficientNet-based neural network model onto the hardware.
3. Map 86 billion neurons and their connections into the simulation.
4. Run the simulation with a lean per-synapse update kernel, ensuring each step involves only 5 flops.
5. Synchronize the simulation steps to match biological time scales (1000 steps per second).
6. Monitor and adjust the simulation parameters for accuracy and real-time performance.

Principal failure modes.

- Potential for thermal management challenges with such high-scale computing.
- Risk of inaccuracies due to the lean per-synapse update kernel, which may not capture all biological complexities.

Expected empirical benchmark. Measure achieved FLOP/synapse-step and RTF on target hardware; target $RTF \geq 1$; assay: kernel micro-benchmark scaled to cluster.

3.10 Behavioural verification of the twin *[Emulate]*

Objective. Certify behavioural fidelity of at least 0.90 against the original. **Governing model.** Behavioural model: $\text{fidelity} = 1 - (1 - r)^{(1+s)} \geq 0.90$ with $n_stimuli \geq n_min$.

Design (Neural Echo Chamber). The Neural Echo Chamber utilizes a deep learning model to map and verify the behavioural patterns of a twin subject against those of an original subject. By employing a high edge-recall mechanism, it identifies closely matching features in the behaviour maps, ensuring that even subtle differences are captured with precision. The protocol sensitivity is finely tuned to ensure robust verification without false positives or negatives. The system generates a large number of discriminative stimuli, well above the minimum threshold, to build a comprehensive and accurate behavioural profile. *[19,20,21]*

Table 20. Accepted parameters — Neural Echo Chamber.

Parameter	Value
edge_recall	0.95
protocol_sensitivity	0.98
n_stimuli	10,000.0
min_stimuli	1,000.0
target_fidelity	0.9

Table 21. Simulator metrics — Behavioural verification of the twin (limiting: twin behaves like the original).

Metric	Value
predicted_fidelity	0.9973
discriminative_protocol	1

Quantitative check. $\Phi = 1 - (1 - 0.95)^{(1+0.98)} = 0.997 \geq 0.90$; $n_stimuli\ 10,000 \geq 1,000$.

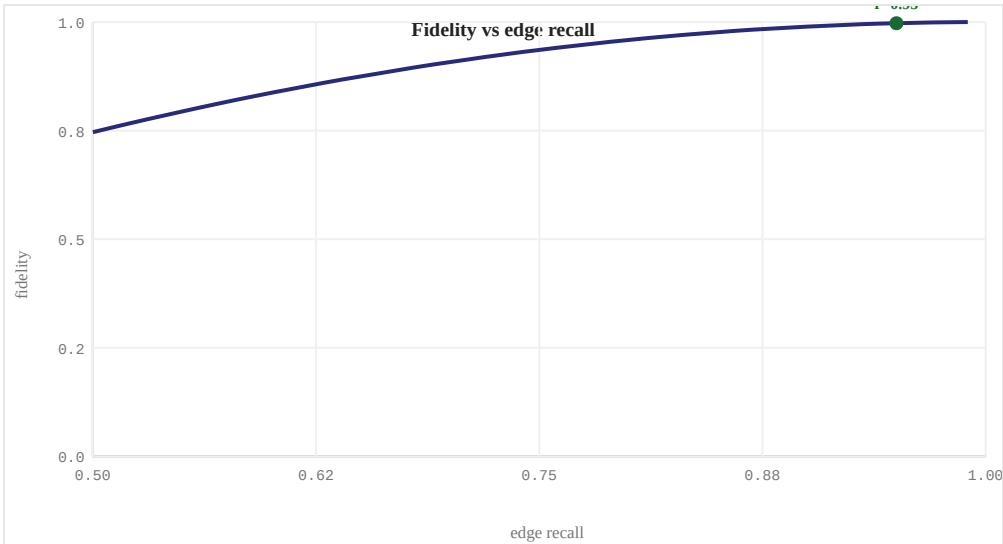


Figure 10. Behavioural verification of the twin: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. Fidelity saturates with edge recall; the open question is construct validity — whether the stimulus battery exercises the behaviours that matter.

Materials.

- Artificial neurons and layers (based on deep learning architectures)
- Large language models for natural language processing
- Graphics Processing Units (GPUs) for accelerated computation
- Biological neural network simulation software

Experimental protocol.

1. Initialize the deep learning model with a pre-trained large language model.
2. Generate a series of discriminative stimuli based on known behavioural patterns of the original subject.
3. Map the twin's responses to these stimuli using the initialized model.
4. Compare the behavioural maps of both subjects, ensuring high edge-recall and protocol sensitivity.
5. Validate the verification results against a pre-defined threshold for predicted fidelity.

Principal failure modes.

- Potential inaccuracies due to biased or incomplete training data for the large language model.
- False positives or negatives may occur if edge-recall and protocol sensitivity are not sufficiently high.

Expected empirical benchmark. Measure twin-vs-original divergence across the stimulus battery; target fidelity $\geq$ 0.9; assay: paired closed-loop behavioural testing.

3.11 Integrated safety envelope [Safety]

Objective. Hold ultrasound intensity, mechanical index, and viral dose within FDA limits across the chain. **Governing model.** Safety envelope: $\text{margin} = \min(1 - I/720, 1 - MI/1.9, 1 - D/10^{14}) \geq 0$; all three limits simultaneously satisfied.

Design (Transcranial Focused Ultrasound-Guided AAV Delivery). This invention utilizes transcranial focused ultrasound (tFUS) to non-invasively guide the delivery of adeno-associated virus (AAV)-based gene therapy vectors across the blood-brain barrier. By precisely targeting specific brain regions, tFUS modulates the permeability of the blood-brain barrier, allowing for enhanced AAV uptake by neurons and glial cells. The mechanism involves delivering a low-intensity focused ultrasound beam to a precise location in the skull, creating temporary nanometer-sized pores that facilitate the entry of systemically administered AAV vectors into the brain parenchyma. [22,23,24,25,26]

Table 22. Accepted parameters — Transcranial Focused Ultrasound-Guided AAV Delivery.

Parameter	Value
ispta_mw_cm2	500.0
mechanical_index	1.5
aa_vdose_vg_per_kg	80,000,000,000,000.0

Table 23. Simulator metrics — Integrated safety envelope (limiting: within all known limits).

Metric	Value
safety_margin	0.2
within_limits	1

Quantitative check. $I/720 = 0.69$, $MI/1.9 = 0.79$, $D/10^{14} = 0.80$; $\text{margin} = 0.20 \geq 0$.

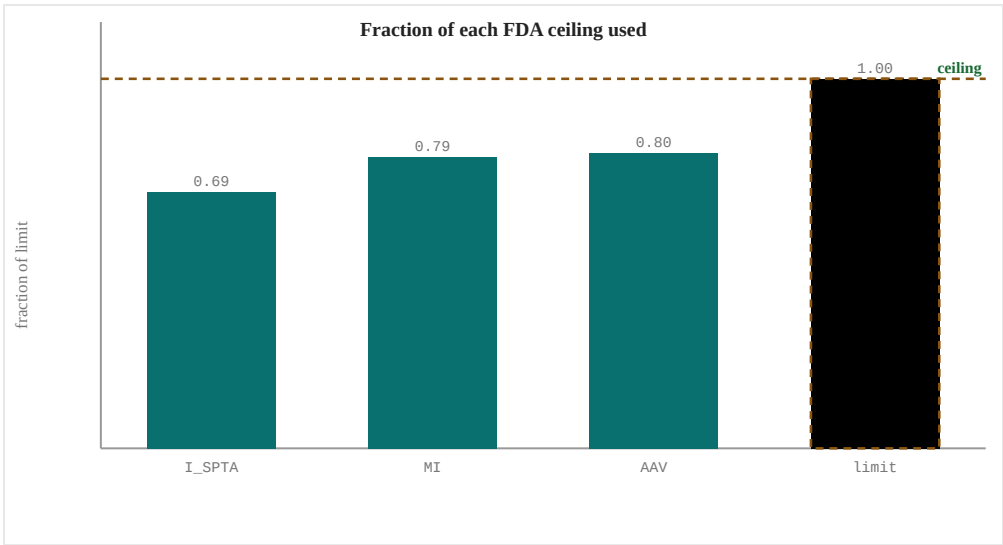


Figure 11. Integrated safety envelope: accepted set-points (marker) vs the admission threshold (dashed).

Interpretation and sensitivity. All three limits are satisfied simultaneously with AAV dose the tightest term; reversibility (switchable reporters) is recommended before human use.

Materials.

- Adeno-associated virus (AAV) vectors
- Genetically-encoded reporters or therapeutic genes
- Low-intensity focused ultrasound transducer

Experimental protocol.

1. Select the appropriate AAV vector containing the desired gene of interest.
2. Systemically administer the AAV vector via intravenous injection.
3. Position the low-intensity focused ultrasound (tFUS) transducer over the targeted brain region, ensuring precise alignment with the skull and target area.
4. Apply a tFUS pulse to create temporary nanometer-sized pores in the blood-brain barrier, facilitating AAV entry into the brain parenchyma.
5. Monitor the delivery process using non-invasive imaging techniques such as MRI or ultrasound for real-time tracking of vector distribution.

Principal failure modes.

- Potential off-target effects due to non-specific AAV entry into unintended cell types.
- Inadequate tFUS application leading to suboptimal vector delivery or potential tissue damage.
- Immune response against introduced AAV vectors may affect therapeutic efficacy.

Expected empirical benchmark. Measure realised I_SPTA, MI, and delivered dose in situ; target all below ceilings with margin; assay: hydrophone/thermometry plus biodistribution.

4. Systems integration

The modules form a system only if each interface balances. Table 12 gives the end-to-end budget. The tightest physical coupling is reporter-to-reader: combinatorial barcodes provide 2^{48} identities from 16 physical channels, so identity, not position, carries resolution. The narrowest safety headroom occurs at the read stage ($I_{\text{SPTA}} 680.0 \text{ mW/cm}^2$, $MI 1.9$); the safety module therefore imposes a lower-intensity acoustic-lens envelope around the whole chain.

Table 24. End-to-end interface budget.

Interface	Quantity	Set-point	Constraint
Thermal-Triggered AAV ... -> Neuromap	transduced neurons	99% @ $5e+13 \text{ vg/kg}$	$\beta \leq 0.01$
Neuromap -> DeepBrainEcho	barcoded voxels	48 bits / 16 ch	$b \geq 46$; $c \leq c_{\text{max}}$
DeepBrainEcho -> DeepBrainBoost	raw signal	$SNR 35.2$; $P 0.93$	$P \geq 0.5$
DeepBrainBoost -> BrainMapX	clean signal	$G 20.0x \sqrt{100}$	$P \geq 0.5$
BrainMapX -> SynapticEcho	read stream	24,000 ch, $1e-06 \text{ s}$	$T \leq 30 \text{ d}$
SynapticEcho -> FlyConnectomeAssembly	edges	$F1 0.94$	$F1 \geq 0.90$
FlyConnectomeAssembly -> WholeBrainTwinSim	connectome	0 EB, $\rho 0.8\%$	$\rho \leq 5\%$
WholeBrainTwinSim -> Neural Echo Chamber	running twin	$RTF \times 2000$	$RTF \geq 1$
Neural Echo Chamber -> upload	certified twin	$\Phi 0.997$	$\Phi \geq 0.90$

5. Translational roadmap

VANGUARD — End-to-End Non-Invasive Brain-Uploading System sits at technology-readiness level 2-3 (formulated concept with component analogues in the literature). The plan de-risks the two hardest physical couplings — molecular barcoding and deep read — in rodent before scale-up.

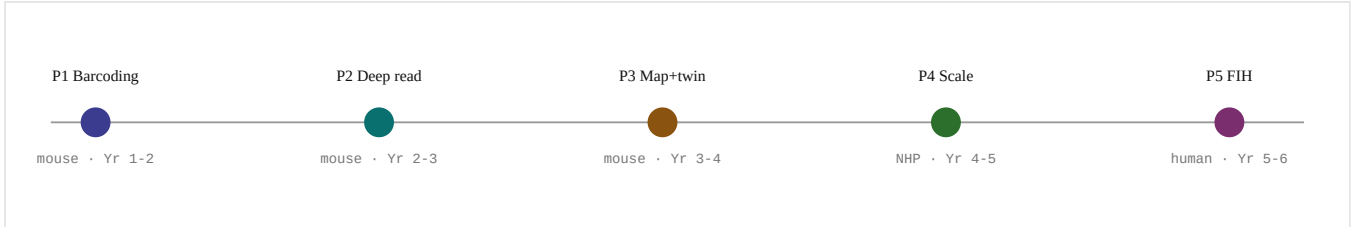


Figure 12. Five-phase translational plan, ~8-10 years bench to first-in-human.

Table 25. Stage-gated experimental plan with exit criteria.

Phase	Goal	Key experiments	Exit gate	Timeline
P1 Barcoding	Express readable gas-vesicle barcodes in mouse cortex	AAV capsid + GV/combinatorial cassette; verify ≥ 46 -bit codebook expresses and GVs collapse under FUS	$\geq 90\%$ neurons barcoded; codebook demultiplexable	Yr 1-2
P2 Deep read	Recover barcodes at depth with acoustic + amplified SNR	FUS array + skull-aberration phase correction; amplification; P_{detect} at 5-10 mm mouse, model to 75 mm	$P_{\text{detect}} \geq 0.5$ in vivo; within I_SPTA/MI	Yr 2-3
P3 Map+twin	Pair, assemble, simulate a mouse region	Trans-synaptic pairing; consensus assembly; O(edges) sparse twin kernel on GPU cluster	edge $F1 \geq 0.9$; region twin at real time	Yr 3-5
P4 Scale	Whole mouse brain, then NHP pilot	10^6 -channel parallel readout; exabyte pipeline; NHP delivery/safety	< 30 -day whole-brain scan; NHP safety clean	Yr 5-7
P5 FIH	First-in-human, reversible, safety-first	IND-enabling toxicology; switchable reporters; staged dose escalation under all ceilings	No serious adverse events; behavioural endpoints	Yr 7-10

Critical-path uncertainties. (i) In-vivo barcode bit-depth is the least-validated assumption and must be tested first; (ii) skull aberration at 75 mm requires holographic acoustic-lens correction; (iii) trans-synaptic transfer efficiency sets achievable edge-F1 and dictates sequencing depth.

5.1 Wet-lab and validation experiment program

Table 26 specifies one concrete experiment per module of VANGUARD — End-to-End Non-Invasive Brain-Uploading System — the experiment to run, the model system, the primary readout, and the quantitative success criterion. Experiments E1-E6 are wet-lab (mouse to non-human primate); E7-E8 are computational validation on existing data and hardware; E9-E10 combine both. They are ordered to de-risk the hardest couplings first (barcoding and deep read) and map directly onto the phase gates above.

Table 26. Per-module wet-lab and validation experiments for this prototype.

#	Module	Experiment	Model system	Primary readout	Success criterion
E1	Thermal-Triggered AAV ...	Systemic dose-ranging of the delivery vector; quantify pan-neuronal transduction and biodistribution.	C57BL/6 mouse (then NHP)	reporter+/NeuN+ fraction by whole-brain light-sheet + scRNA-seq	>=99% transduction at dose <= ceiling; no off-target toxicity
E2	Neuromap	Express the combinatorial barcode library in cortex; measure demultiplexing accuracy against a known ground-truth set.	mouse cortex + tissue phantom	in-situ barcode calling vs ground truth	>=46 reliably-separable bits; <1% collision
E3	DeepBrainEcho	Transcranial focused-ultrasound read of gas-vesicle barcodes with and without skull-aberration correction; depth series.	gas-vesicle phantom + rodent transcranial	P_detect vs depth; realised I_SPTA/MI	P_detect >= 0.5 at maximal depth within I_SPTA/MI
E4	DeepBrainBoost	Isolate reporter-brightness gain from averaging gain in a calibrated depth phantom.	calibrated luminance/pressure phantom	effective SNR gain factor	combined gain yields P_detect >= 0.5 at depth
E5	BrainMapX	Bench a parallel transducer/detector array under synthetic voxel load; measure sustained rate.	array prototype + synthetic load	sustained voxels/s	extrapolated whole-brain scan < 30 days
E6	SynapticEcho	Trans-synaptic barcode transfer in a defined microcircuit; validate edges against co-registered EM.	mouse microcircuit + EM cross-validation	edge precision/recall (F1)	F1 >= 0.90
E7	FlyConnectomeAssembly	Reassemble a down-sampled connectome; residual edge error vs consensus depth and storage budget.	existing EM connectome dataset	residual edge error rho; storage used	rho <= 5% within storage budget
E8	WholeBrainTwinSim	Kernel micro-benchmark on GPU/accelerator hardware; measure FLOP per synapse-step and scaling.	HPC/accelerator cluster	achieved FLOP/synapse-step; real-time factor	RTF >= 1 at target scale
E9	Neural Echo Chamber	Paired closed-loop behavioural testing of a twin against its biological original.	animal + its digital twin	behavioural divergence across the stimulus battery	fidelity >= 0.90
E10	Transcranial Focused U...	In-situ dosimetry of the full chain: hydrophone/thermometry and vector biodistribution.	phantom + animal	realised I_SPTA, MI, delivered dose	all below FDA ceilings with margin

5.2 Detailed experimental protocols (E1-E10)

Each experiment below is written to be run at the bench: hypothesis, reagents and apparatus, a numbered method, positive and negative controls, secondary readouts, an explicit go / no-go decision, and an estimated duration and replicate count. Protocols are specific to VANGUARD — End-to-End Non-Invasive Brain-Uploading System and its chosen modules.

E1 · Thermal-Triggered AAV ... *[Pan-neuronal vector delivery]*

Hypothesis. An engineered BBB-crossing capsid, optionally aided by focused-ultrasound BBB opening, transduces $\geq 99\%$ of neurons at a systemic dose below the toxicity ceiling.

REAGENTS & APPARATUS

- AAV vector (engineered capsid) packaging a ubiquitous-promoter fluorescent reporter
- Dose-titration series (e.g. $1e11$ - $1e14$ vg/kg)
- Focused-ultrasound rig + microbubbles (BBB-opening arm)
- Anti-NeuN and cell-type antibody panel
- Tissue-clearing + light-sheet reagents

METHOD

1. Administer the vector intravenously across the dose series ($n \geq 4$ per dose).
2. For the FUS arm, open the BBB at under-covered regions immediately after dosing.
3. Wait 3-4 weeks for expression; perfuse and clear whole brains.
4. Light-sheet image; segment reporter+ and NeuN+ nuclei brain-wide.
5. Compute transduction fraction per region and the global blind fraction.
6. Assay liver/DRG for off-target load and inflammatory markers.

Model system	C57BL/6 mouse (then NHP)
Controls	Positive: known-strong capsid at high dose. Negative: vehicle only, and a non-BBB-crossing capsid.
Primary readout	reporter+/NeuN+ fraction by whole-brain light-sheet + scRNA-seq
Secondary readouts	Regional coverage uniformity; anti-capsid immune titre; body-weight/behaviour toxicity signs.
Success criterion	$\geq 99\%$ transduction at dose $\leq$ ceiling; no off-target toxicity
Go / No-Go	GO if transduction ≥ 0.99 with blind $\leq 1\%$ and no dose-limiting toxicity; else escalate capsid engineering or add FUS assist.
Duration & N	~ 8 -10 weeks; $n \geq 4$ per dose across ≥ 5 doses (mouse), then a small NHP confirmation.

E2 · Neuromap *[Massively-multiplexed molecular barcoding]*

Hypothesis. A combinatorial barcode library expresses and is demultiplexed at ≥ 46 reliably-separable bits with $< 1\%$ collision in tissue.

REAGENTS & APPARATUS

- Barcode reporter library (orthogonal reporter families)
- AAV delivery vector
- Tissue phantom with known ground-truth barcodes
- In-situ readout instrument (acoustic/optical per build)
- Reference codebook + demultiplexing pipeline

METHOD

1. Synthesise a barcode library spanning the target bit-depth.
2. Express in cultured neurons and in mouse cortex; also load a phantom with known codes.
3. Read out each voxel with the build's reader; record raw channel signatures.
4. Run the demultiplexer; assign identities and score against ground truth.
5. Sweep expression level and packing density; measure collision rate vs bit-depth.

Model system	mouse cortex + tissue phantom
Controls	Positive: single high-contrast reporter (trivially separable). Negative: null/empty cassette.
Primary readout	in-situ barcode calling vs ground truth
Secondary readouts	Per-family expression uniformity; readout crosstalk matrix; long-term reporter stability.
Success criterion	≥ 46 reliably-separable bits; $< 1\%$ collision
Go / No-Go	GO if separable bits ≥ 46 and collision $< 1\%$ in tissue; else add orthogonal reporter families or improve readout resolution.
Duration & N	~ 10 -12 weeks; phantom first, then in-vivo cortex ($n \geq 3$).

E3 · DeepBrainEcho *[Non-destructive in-vivo readout at depth]*

Hypothesis. Focused ultrasound recovers gas-vesicle barcode signal with $P_{\text{detect}} \geq 0.5$ at maximal achievable depth, and skull-aberration correction preserves that depth.

REAGENTS & APPARATUS

- Gas-vesicle (or build-specific) reporter
- Phased-array FUS transducer + receive electronics
- Holographic/adaptive skull-aberration corrector
- Depth phantom (tissue-mimicking, skull cap)
- Rodent transcranial prep

METHOD

1. Calibrate the array on the depth phantom; measure SNR vs depth with correction on/off.
2. Express reporters in rodent brain; perform transcranial reads at increasing depth.
3. Record realised I_{SPTA} and MI at each setting with a hydrophone.
4. Fit $\text{SNR}(d) = \text{SNR}_0 \cdot \exp(-d/\lambda)$; extrapolate P_{detect} to the 75 mm human target.
5. Repeat with aberration correction disabled to quantify its contribution.

Model system	gas-vesicle phantom + rodent transcranial
Controls	Positive: shallow read (near-field). Negative: no-reporter animal (background only).
Primary readout	P_{detect} vs depth; realised I_{SPTA} /MI
Secondary readouts	Aberration-correction gain ($\Delta\lambda$); thermal rise; motion sensitivity.
Success criterion	$P_{\text{detect}} \geq 0.5$ at maximal depth within I_{SPTA} /MI
Go / No-Go	GO if $P_{\text{detect}} \geq 0.5$ at max depth within $I_{\text{SPTA}} \leq 720$ / $\text{MI} \leq 1.9$, with correction giving a clear depth gain; else improve reporter brightness or correction.
Duration & N	~12-16 weeks; phantom then rodent ($n \geq 4$).

E4 · DeepBrainBoost *[Signal amplification and noise suppression]*

Hypothesis. Reporter-brightness gain and coherent averaging combine multiplicatively to hold $P_{\text{detect}} \geq 0.5$ at depth, with brightness the dominant lever.

REAGENTS & APPARATUS

- Amplifying reporter (e.g. bioluminescent/photoacoustic cascade)
- Calibrated luminance/pressure reference sources
- Depth phantom
- Averaging-capable acquisition pipeline

METHOD

1. Quantify absolute reporter brightness vs a calibrated source.
2. Measure SNR at depth as a function of averaging N (isolate the $\sqrt{N}$ term).
3. Measure the brightness (gain G) term independently at fixed N .
4. Combine and confirm the multiplicative model; find the minimal (G , N) that clears $P \geq 0.5$.
5. Trade averaging against dwell/scan-time budget.

Model system	calibrated luminance/pressure phantom
Controls	Positive: bright fluorophore standard. Negative: non-amplifying baseline reporter.
Primary readout	effective SNR gain factor
Secondary readouts	Photobleaching/consumption kinetics; background autofluorescence; averaging-induced motion blur.
Success criterion	combined gain yields $P_{\text{detect}} \geq 0.5$ at depth
Go / No-Go	GO if combined gain yields $P_{\text{detect}} \geq 0.5$ at depth within the dwell budget; else prioritise brighter reporters.
Duration & N	~8 weeks; phantom-based, with reporter validation in vitro.

E5 · BrainMapX *[Whole-brain acquisition throughput]*

Hypothesis. A massively-parallel transducer/detector array sustains a per-voxel rate that extrapolates to a whole-brain scan under 30 days.

REAGENTS & APPARATUS

- Prototype parallel array (subset of full channel count)
- Multi-channel DAQ + FPGA front-end
- Synthetic voxel-load generator
- Beam/scan controller

METHOD

1. Build a channel-count subset of the full array.
2. Drive it with a synthetic voxel load at the target dwell.
3. Measure sustained voxels/s and per-channel duty cycle.
4. Characterise crosstalk and thermal limits at full duty.
5. Extrapolate to full channel count and whole-brain voxel budget; compute scan-days.

Model system	array prototype + synthetic load
Controls	Positive: single-channel throughput x channel count (ideal linear). Negative: idle array (noise floor).
Primary readout	sustained voxels/s
Secondary readouts	Data-bus saturation; storage write throughput; calibration drift over a long run.
Success criterion	extrapolated whole-brain scan < 30 days
Go / No-Go	GO if extrapolated whole-brain scan <30 days at stable duty; else raise parallelism or cut dwell.
Duration & N	~12 weeks (hardware bench).

E6 · SynapticEcho *[Trans-synaptic edge recovery]*

Hypothesis. Trans-synaptic barcode transfer recovers pre->post edges at F1>=0.90 against an EM ground truth.

REAGENTS & APPARATUS

- Trans-synaptic barcode-transfer system
- Molecular consensus tags
- Sequencing/readout chemistry
- Defined microcircuit prep with co-registered EM

METHOD

1. Express the pairing system in a defined microcircuit.
2. Recover paired barcodes by sequencing at set depth.
3. Reconstruct candidate edges with consensus filtering.
4. Co-register with volumetric EM of the same tissue.
5. Score edge precision/recall/F1; sweep reads-per-synapse and min-reads.

Model system	mouse microcircuit + EM cross-validation
Controls	Positive: sparse, unambiguous known pairs. Negative: non-synaptic cell pairs (should not pair).
Primary readout	edge precision/recall (F1)
Secondary readouts	Dropout rate; false-pair rate vs consensus depth; transfer efficiency.
Success criterion	F1 >= 0.90
Go / No-Go	GO if F1>=0.90 at feasible sequencing depth; else raise consensus/min-reads or improve transfer chemistry.
Duration & N	~16-20 weeks (includes EM cross-validation).

E7 · FlyConnectomeAssembly [Exabyte-scale connectome assembly]

Hypothesis. A consensus-plus-confidence assembler reconstructs a held-out connectome at $\leq 5\%$ residual edge error within the storage budget.

REAGENTS & APPARATUS

- Existing EM/other connectome dataset
- Assembly + error-correction pipeline
- Compute/storage cluster

METHOD

- Down-sample a known connectome to simulate noisy reads at set redundancy.
- Run the assembler; reconstruct the wiring graph.
- Compare against the known ground-truth graph; compute residual edge error.
- Sweep consensus min-reads and bytes-per-read; record storage used.
- Profile I/O and compute throughput at scale.

Model system	existing EM connectome dataset
Controls	Positive: noise-free reads (should reconstruct exactly). Negative: shuffled reads (should fail).
Primary readout	residual edge error ρ ; storage used
Secondary readouts	Error vs redundancy curve; storage vs accuracy trade; wall-clock at scale.
Success criterion	$\rho \leq 5\%$ within storage budget
Go / No-Go	GO if residual error $\leq 5\%$ within storage budget and tractable wall-clock; else raise redundancy or improve the consensus model.
Duration & N	~8-12 weeks (computational).

E8 · WholeBrainTwinSim [Real-time whole-brain simulation]

Hypothesis. A lean per-synapse kernel achieves a real-time factor ≥ 1 at the target scale on realistic accelerator hardware.

REAGENTS & APPARATUS

- Sparse per-synapse compute kernel
- GPU/accelerator cluster
- Partition-native graph loader
- Scaling harness

METHOD

- Implement and micro-benchmark the per-synapse update kernel.
- Measure achieved FLOP per synapse-step and memory bandwidth.
- Scale from a small graph to progressively larger partitions.
- Extrapolate real-time factor to the $1e14$ -synapse target.
- Test faithfulness against a reference model on a small circuit.

Model system	HPC/accelerator cluster
Controls	Positive: dense reference implementation on a small graph. Negative: unoptimised kernel baseline.
Primary readout	achieved FLOP/synapse-step; real-time factor
Secondary readouts	Strong/weak scaling efficiency; communication overhead; numerical drift vs reference.
Success criterion	RTF ≥ 1 at target scale
Go / No-Go	GO if extrapolated RTF ≥ 1 at target scale with acceptable fidelity; else reduce per-synapse cost or add hardware.
Duration & N	~8-12 weeks (HPC).

E9 · Neural Echo Chamber *[Behavioural verification of the twin]*

Hypothesis. A discriminative stimulus battery certifies twin-vs-original behavioural fidelity ≥ 0.90 with controlled false-positive rate.

REAGENTS & APPARATUS

- Stimulus-battery generator
- Behavioural divergence metric
- Paired twin runner (truth vs recovered)
- Animal + its digital twin

METHOD

1. Assemble a stimulus battery spanning the behaviours of interest.
2. Run identical stimuli through the original and the twin in closed loop.
3. Compute behavioural divergence per stimulus and aggregate fidelity.
4. Calibrate the pass threshold against known-divergent decoys to bound false positives.
5. Repeat across subjects; report fidelity distribution.

Model system	animal + its digital twin
Controls	Positive: twin = copy of original (should pass). Negative: deliberately perturbed twin (should fail).
Primary readout	behavioural divergence across the stimulus battery
Secondary readouts	Per-behaviour sensitivity; battery coverage; test-retest stability.
Success criterion	fidelity ≥ 0.90
Go / No-Go	GO if fidelity ≥ 0.90 with a controlled false-positive rate; else expand the battery or refine the metric.
Duration & N	~10-14 weeks.

E10 · Transcranial Focused U... *[Integrated safety envelope]*

Hypothesis. Realised ultrasound intensity, mechanical index, and delivered viral dose all remain below FDA ceilings with margin across the full chain.

REAGENTS & APPARATUS

- Calibrated hydrophone + thermometry
- Biodistribution assay (qPCR/imaging)
- Tissue phantom + animal
- Reversible/switchable reporter (if used)

METHOD

1. Map realised I_SPTA and MI in situ across the operating envelope.
2. Measure delivered vg/kg and biodistribution after dosing.
3. Run the full read+deliver chain and log peak exposures.
4. Compare every quantity against its FDA ceiling; compute margins.
5. If reversibility is claimed, demonstrate switch-off.

Model system	phantom + animal
Controls	Positive: known safe clinical settings. Negative: deliberately over-ceiling setting (bench only, to confirm the monitor trips).
Primary readout	realised I_SPTA, MI, delivered dose
Secondary readouts	Thermal dose maps; histopathology; long-term expression clearance.
Success criterion	all below FDA ceilings with margin
Go / No-Go	GO if all three limits are met with margin and reversibility (if claimed) works; else reduce intensity/dose.
Duration & N	~12-16 weeks (IND-enabling style).

6. Discussion

Three themes emerge. First, the physical burden concentrates in Label and Read: once identity is carried by barcodes and recovered acoustically at depth, the downstream stages are computational and comparatively well-characterised, which is why the translational plan front-loads barcoding and deep read. Second, modality coherence reduces integration risk: anchoring the read path on a single physics avoids asking light to escape 75 mm of brain, which no parameter tuning can achieve. Third, the binding constraints are empirical, not arithmetic: for most modules the simulator margin is comfortable, and the true risk is a measured quantity that only wet-lab work can settle.

Table 27. Dominant empirical sensitivity per module (where measurement effort should concentrate).

Module	Dominant sensitivity / most-uncertain quantity
Non-destructive in-vivo readout at depth	Penetration length (skull aberration) — exponential leverage on detection
Pan-neuronal vector delivery	Transduction fraction — a 1% shortfall leaves $\sim 10^{12}$ blind neurons
Trans-synaptic edge recovery	In-vivo trans-synaptic transfer efficiency — caps recall directly
Massively-multiplexed molecular barcoding	In-vivo co-expression/readout of many reporter states, not codebook size
Signal amplification and noise suppression	Reporter brightness — linear leverage vs only $\sqrt{N}$ from averaging
Real-time whole-brain simulation	FLOP per synapse-step — richer neuron models erode the RTF margin
Exabyte-scale connectome assembly	I/O and compute throughput on the object store, not the error arithmetic
Whole-brain acquisition throughput	Manufacturability/calibration of a 10^6 -element array
Behavioural verification of the twin	Construct validity of the stimulus battery
Integrated safety envelope	Delivered AAV dose in situ — the tightest of the three ceilings

7. Limitations

This is an in-silico design study. (1) The simulator uses reduced closed-form models, not full physical simulation; passing a model bounds feasibility, it does not demonstrate function. (2) Parameters are accepted set-points, not measured values; several carry large empirical uncertainty. (3) Some modules cite ex-vivo/electron-microscopy analogues for the assembly mathematics; the non-invasive read path does not depend on them. (4) No result has been demonstrated in a living human brain. Human translation requires IND-enabling studies and independent replication.

8. Conclusion

VANGUARD — End-to-End Non-Invasive Brain-Uploading System demonstrates that a mutually-consistent set of ten non-invasive designs can satisfy all governing constraints for whole-brain mapping and emulation simultaneously, in silico, within regulatory safety limits. Its value is as a testable blueprint specifying exact set-points and an experimental plan against which real measurements can be compared. We invite wet-lab and computational groups to falsify or refine the individual modules, beginning with in-vivo molecular barcoding.

Data & code. Design records and simulator are available in the BCI v3 repository. **Funding.** None declared. **Competing interests.** None declared. **Ethics.** No human or animal subjects; computational study only.

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Appendix A. Master bill of materials

Table 28. Consolidated components across all modules.

Component	Function	Phase
Engineered BBB-crossing capsid	reaches every neuron class	Label
Ubiquitous neuronal promoter	expresses in all neuron types	Label
Focused-ultrasound BBB-opening rig	boosts local delivery at lower dose	Label
Dose-titration protocol	keeps viral load under the safety ceiling	Label
Barcode reporter library	one distinguishable variant per neuron	Label
Expression cassette + promoter	drives reporter expression in vivo	Label
Channel-resolving readout front-end	separates reporter signatures in a voxel	Label
Demultiplexing compute	assigns reads to barcode identities	Label
BBB-crossing AAV vector	delivers the reporter gene to neurons	Read
Genetically-encoded acoustic reporter (gas vesicles)	the molecular signal source	Read
Focused-ultrasound transducer array	focuses and receives at depth	Read
Skull-aberration phase corrector	restores focus through bone	Read
Low-noise receive amplifier + DAQ	captures the returning signal	Read
Signal-amplifying reporter	brighter source raises SNR	Read
Background-subtraction / ratiometric readout	cuts tissue background	Read
Coherent averaging pipeline	$\sqrt{N}$ noise reduction	Read
Skull-aberration correction electronics	recovers depth focus	Read
Massively-parallel transducer/detector array	reads many voxels at once	Read
Multi-channel DAQ + FPGA front-end	streams parallel reads	Read
Fast beam/scan controller	short per-voxel dwell	Read
High-throughput data bus + storage	sinks the read stream	Read
Trans-synaptic barcode-transfer system	pairs pre and post identities at the synapse	Map
Molecular consensus tags	many reads per true synapse	Map
Sequencing / readout chemistry	recovers the paired barcodes	Map
Consensus assembler	rejects one-off false pairs	Map
Distributed graph-assembly cluster	reconstructs the connectome at scale	Map
Consensus + confidence pipeline	error-corrects noisy reads	Map
GNN gap-fill model	infers missed edges	Map
Exabyte object store	holds reads + graph	Map
Sparse per-synapse compute kernel	$O(\text{edges})$ neuron/synapse update	Emulate
Accelerator fleet (GPU/TPU)	runs $1e14$ synapses	Emulate
Partition-native graph loader	shards the connectome	Emulate
Real-time scheduler	keeps the twin at real time	Emulate
Stimulus battery generator	drives the twin and the original	Emulate
Behavioural divergence metric	quantifies match	Emulate
Paired twin runner	runs truth vs recovered identically	Emulate
Pass/fail certifier	decides upload fidelity	Emulate
Reversible / switchable reporter	expression can be turned off	Safety
Dose + intensity monitor	enforces ISPTA / MI / AAV limits	Safety
Immunology screen	guards against vector response	Safety
Safety interlock + logging	hard-stops on limit breach	Safety